

Part B

Q.1) Hotel Banking System

→ Algorithm

- Take input for number of guests
- Take season and room type input
- Calculate accordingly to the rates

Peak Szn → - Standard: 5,000

- Deluxe: 8,000

- Suite: 12,000

Off-Peak Szn → - Standard: 3,000

- Deluxe: 5,000

- Suite: 8,000

• If nights > 7 → 15% discount → Total to hotel revenue

→ Problem Analysis Chart & IPO chart

Input	Process	Output
N(no. of guests)	- Loop N times, rate based on Szn + room type	Final price per guest
Season	- total = rate × nights	
Room type	- night > 7 - 15% off	
Nights stayed	- Total → Hotel Revenue	Hotel Revenue

Pseudocode

START

total $\leftarrow$ 0

cost $\leftarrow$ 0

INPUT N

FOR i $\leftarrow$ 1 TO N

INPUT season

INPUT room-type

INPUT night-stayed

IF season = "Peak" THEN

IF room-type == "standard" THEN

cost $\leftarrow$ 5,000 * night-stayed

Else IF room-type == "Deluxe" THEN

cost $\leftarrow$ 8,000 * night-stayed

Else:

cost $\leftarrow$ 12,000 * night-stayed

ENDIF

ELSE:

IF room-type == "standard" THEN

cost $\leftarrow$ 3,000 * night-stayed

Else IF room-type == "Deluxe" THEN

cost $\leftarrow$ 5,000 * night-stayed

Else:

cost $\leftarrow$ 8,000 * night-stayed

ENDIF

ENDIF

IF high-stayed > 7 THEN

~~cost~~ temp = cost * 0.15

cost ← cost - temp

Else:

ENDIF

Total ← total + cost

NEXT

OUTPUT " Total hotel revenue ", Total

END

Flowchart

if night < 10 then

cost = 1200 * night

total = total + cost

